

# PRISMA CRASH COURSE



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## Prisma ORM in One Video 🔥 Setup, Migrations, CRUD & Best Practices

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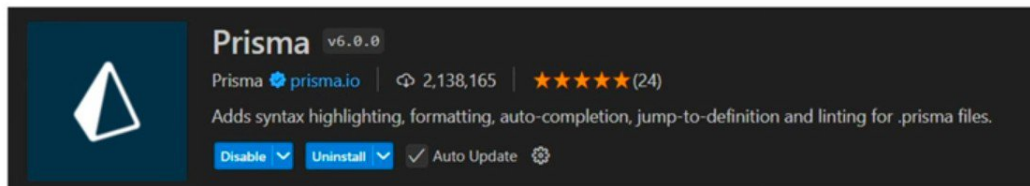
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आता लगेच जॉइन करा!

# Introduction to Prisma ORM

- Prisma is an open-source ORM (Object-Relational Mapping) tool for Node.js and TypeScript that simplifies database interaction.
- It helps you interact with databases using a type-safe query builder and eliminates the need to write raw SQL queries.
- Prisma provides Prisma Client, a powerful auto-generated query builder, and Prisma Migrate for database schema migrations.
- <https://www.prisma.io/docs/getting-started/setup-prisma>
- Install **Prisma** Vscode Extension





## Prisma Client

Auto-generated  
and type-safe  
database client.



## Prisma Migrate

Declarative data modeling  
and customizable  
migrations.



## Prisma Studio

Modern UI to view  
and edit your  
application data.

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From the CLI

Import from existing database

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5 Min

Set up Prisma ORM

Start from scratch or add Prisma ORM to an existing project. The following tutorials introduce you to the [Prisma CLI](#), [Prisma Client](#), and [Prisma Migrate](#).

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Relational databases

Learn how to create a new TypeScript project with a Prisma Postgres database from scratch. This tutorial introduces you to the [Prisma CLI](#), [Prisma Client](#), and [Prisma Migrate](#) and covers the following workflows:

Creating a TypeScript project on your local machine from scratch

Creating a [Prisma Postgres](#) database

Schema migrations and queries (via [Prisma ORM](#))

Connection pooling and caching (via [Prisma Accelerate](#))

Real-time database change events (via [Prisma Pulse](#))

Prerequisites #

To successfully complete this tutorial, you need:

a [Prisma Data Platform](#) (PDP) account

[Node.js](#) installed on your machine (see [system requirements](#) for officially supported versions) (see [system requirements](#) for officially supported versions)

Create project setup

Create a project directory and navigate into it:

```
$ mkdir hello-prisma
$ cd hello-prisma
```

Next, initialize a TypeScript project and add the Prisma CLI as a development dependency to it:

```
$ npm init -y
```

AI

1:16 PM 2/5/2025

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Create project setup

Create a project directory and navigate into it:

```
$ mkdir hello-prisma
$ cd hello-prisma
```

Next, initialize a TypeScript project and add the Prisma CLI as a development dependency to it:

```
$ npm init -y
$ npm install prisma typescript tsx @types/node --save-dev
```

This creates a `package.json` with an initial setup for your TypeScript app.

Next, initialize TypeScript:

```
$ npx tsc --init
```

You can now invoke the Prisma CLI by prefixing it with `npx`:

```
$ npx prisma
```

Next, set up your Prisma ORM project by creating a `schema.prisma` file in the project root directory using the following command:

```
$ npx prisma init
```

schema.prisma...

prisma\_mysql - Visual...

1:17 PM

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```
3 url = env("DATABASE_URL")
4 }
```

In this case, the `url` is [set via an environment variable](#) which is defined in `.env`:

```
.env
DATABASE_URL="mysql://johndoe:randompassword@localhost:3306/mydb"
```

INFO

We recommend adding `.env` to your `.gitignore` file to prevent committing your environment variables.

You now need to adjust the connection URL to point to your own database.

The [format of the connection URL](#) for your database typically depends on the database you use. For MySQL, it looks as follows (the parts [encased in red](#) are *placeholders* for your specific connection details):

```
mysql://USER:PASSWORD@HOST:PORT/DATABASE
```

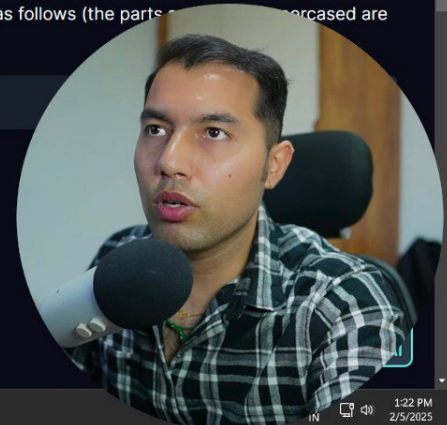
Here's a short explanation of each component:

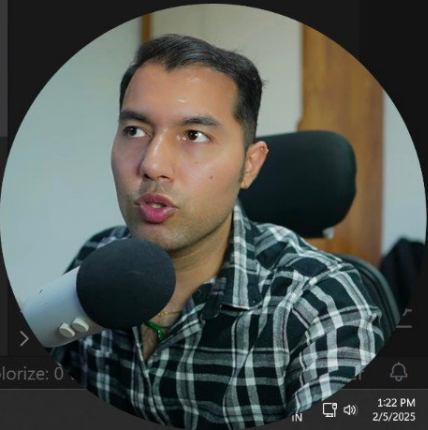
- `USER`: The name of your database user
- `PASSWORD`: The password for your database user
- `PORT`: The port where your database server is running (typically `3306` for MySQL)
- `DATABASE`: The name of the [database](#)

As an example, for a MySQL database hosted on AWS RDS, the [connection URL](#) might look similar to this:

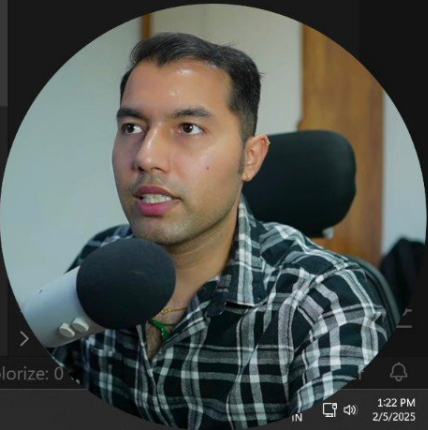
1:22 PM

2/5/2025









schema.prisma - prisma\_mysql - Visual Studio Code

prisma > schema.prisma > db

```
4 // Looking for ways to speed up your queries, or scale
5 // Try Prisma Accelerate: https://pris.ly/cli/accelerate-init
6
7 generator client {
8   provider = "prisma-client-js"
9 }
10
11 datasource db {
12   provider = "mysql"
13   url      = env("DATABASE_URL")
14 }
15
```

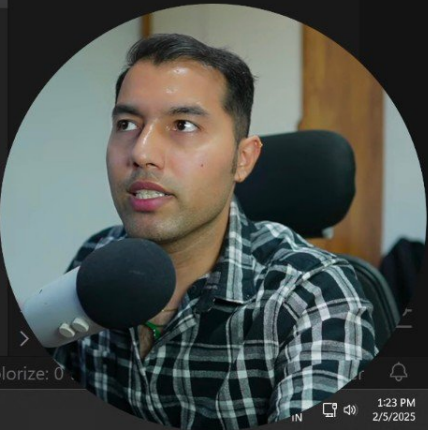
Generate

EXPLORER

- OPEN EDITORS
  - schema.prisma prisma U
- PRISMA\_MYSQL
  - node\_modules
  - prisma
    - schema.prisma U
  - .env
  - .gitignore U
  - package-lock.json U
  - package.json U

Ln 14, Col 2 Spaces: 2 UTF-8 LF {} Prisma Go Live Colorize: 0

29°C Sunny 1:23 PM 2/5/2025



# Migrations in ORM

- As you know that, SQL based databases require you to assign schema which you can't do while backend is running; you have to do beforehand. Migrations is for that.
- Migration is the process of evolving and managing changes to a database schema over time.
- It involves creating, modifying, or deleting tables, columns, and relationships in the database.
- In Prisma, migrations are stored as versioned SQL script files.
- Migrations help ensure that database structures stay consistent across environments (e.g., development, staging, production).
- Prisma Migrate generates and applies migrations automatically based on changes in the Prisma schema.



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Using Prisma Migrate

Creating the database schema

In this guide, you'll use [Prisma Migrate](#) to create the tables in your database. Add the following data model to your [Prisma schema](#) in `prisma/schema.prisma`:

prisma/schema.prisma

```
1 model Post {
2   id      Int      @id @default(autoincrement())
3   createdAt DateTime @default(now())
4   updatedAt DateTime @updatedAt
5   title    String   @db.VarChar(255)
6   content  String?
7   published Boolean  @default(false)
8   author   User      @relation(fields: [authorId], references: [id])
9   authorId Int
10 }
11
12 model Profile {
13   id      Int      @id @default(autoincrement())
14   bio     String?
15   user    User      @relation(fields: [userId], references: [id])
16   userId  Int       @unique
17 }
```

TS TypeScript

PostgreSQL

29°C Sunny

Windows Taskbar

1:24 PM 2/5/2025

Thapa Technical



In this guide, you'll use [Prisma Migrate](#) to create the tables in your database. Add the following data n

📄 prisma/schema.prisma

```
1  model Post {
2    id          Int      @id @default(autoincrement())
3    createdAt   DateTime @default(now())
4    updatedAt   DateTime @updatedAt
5    title       String    @db.VarChar(255)
6    content     String?
7    published   Boolean   @default(false)
8    author      User      @relation(fields: [authorId], references: [id])
9    authorId    Int
10 }
11
12 model Profile {
13   id      Int      @id @default(autoincrement())
14   bio     String?
15   user    User      @relation(fields: [userId], references: [id])
16   userId  Int      @unique
17 }
```



Visual Studio Code interface showing a Prisma schema file and a terminal window.

**Schema File (schema.prisma):**

```
prisma > schema.prisma > db

10
11 datasource db {
12   provider = "mysql"
13   url       = env("DATABASE_URL")

```

**Terminal Window:**

```
node_2025\prisma_project\prisma_mysql on HEAD (bbf3403) [?!] is v1.0.0 via v23.2.0
o) npx prisma migrate dev --name init
```

**Bottom Bar:**

Ln 14, Col 2 Spaces: 2 UTF-8 LF {} Prisma Go Live Colorize: 0

Breaking news PM Modi arrives...

1:30 PM 2/5/2025

**Avatar:** A circular profile picture of a man with short dark hair, wearing a plaid shirt, speaking into a microphone.

**Logo:** Shupn Technical

# Create ShortLink model

```
model ShortLink {  
  id          Int    @id @default(autoincrement())  
  shortCode   String @unique  
  url         String  
  createdAt   DateTime @default(now())  
  updatedAt   DateTime @updatedAt  
}
```

## Migrate it

- Now, run **npx prisma migrate dev** to create migration files and push it.
- You can pass **--name** flag with it to include name:
  - **npx prisma migrate dev --name init**
- In Production, you can run **npx prisma migrate deploy** to run migration files.
- You can also include both of them in **package.json** scripts



Visual Studio Code interface showing a Prisma schema file named `schema.prisma` in the `prisma_mysql` project.

The schema defines a `datasource db` and a `model User`.

```
11  datasource db {
12    provider = "mysql"
13    url      = env("DATABASE_URL")
14  }
15
16  model User {
17    id      Int      @id @default(autoincrement())
18    email   String   @unique
19    name    String?
20    createdAt DateTime @default(now())
21    updatedAt DateTime @default(now()) @updatedAt
22  }
23
```

The status bar at the bottom indicates the file is at Line 21, Column 48, using UTF-8 encoding and LF line endings. The system tray shows the temperature is 29°C and it is sunny.





Visual Studio Code interface showing the editing of `package.json` for a project named `prisma_mysql`.

The Explorer sidebar on the right shows the project structure:

- PRISMA\_MYSQL
  - node\_modules
  - prisma
    - migrations
      - schema.prisma
  - .env
  - .gitignore
  - package-lock.json
  - package.json

The main editor displays the `package.json` file with the following content:

```
10  "license": "ISC",
11  "description": "",
12  "devDependencies": {
13    "prisma": "^6.3.1"
14  },
15  "dependencies": {
16    "@prisma/client": "^6.3.1"
17  }
18 }
19
```

The status bar at the bottom indicates the current position is Line 16, Column 31, with 2 spaces, UTF-8 encoding, LF line endings, and a JSON file type. It also shows the Go Live extension and a Colorize setting of 0.

A circular video feed in the bottom right corner shows a man speaking into a microphone.

136 PM 2/5/2025

MySQL Workbench

Local instance MySQL80 x

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- mysql\_users
- prismaql**
- sys
- url\_shortener
- url\_shortener\_mysql
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Schema: **prismaql**

Object Info Session

SQL File 1\*

Limit to 1000 rows

```
1 select * from user;
```

Result Grid

| id | email | name | createdAt | updatedAt |
|----|-------|------|-----------|-----------|
| *  |       |      |           |           |

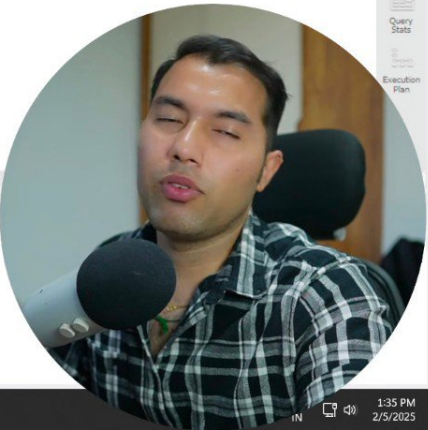
user 1 x

Output

Action Output

| # | Time     | Action                           | Message           |
|---|----------|----------------------------------|-------------------|
| 1 | 13:35:17 | select * from user LIMIT 0, 1000 | 0 row(s) returned |

1:35 PM 2/5/2025



Visual Studio Code interface showing a migration file named `migration.sql` in the `prisma_mysql` workspace.

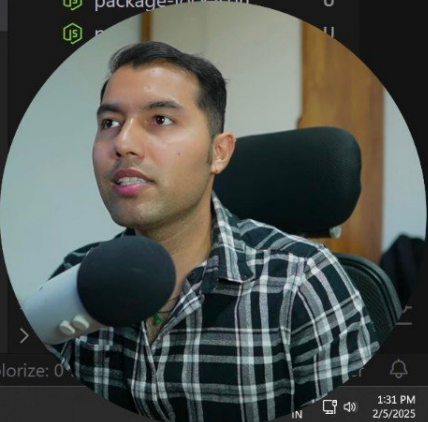
The Explorer sidebar on the right shows the file structure:

- prisma > migrations > 20250205080058\_init > migration.sql
- Other files: .env, schema.prisma, migration\_lock.toml, .gitignore, package-lock.json.

The main editor displays the SQL migration code:

```
1  -- CreateTable
2  CREATE TABLE `User` (
3      `id` INTEGER NOT NULL AUTO_INCREMENT,
4      `email` VARCHAR(191) NOT NULL,
5      `name` VARCHAR(191) NULL,
6      `createdAt` DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
7      `updatedAt` DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
8
9      UNIQUE INDEX `User_email_key`(`email`),
10     PRIMARY KEY (`id`)
11 ) DEFAULT CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
12
```

The status bar at the bottom indicates the current cursor position is `Ln 1, Col 1` with `Spaces: 4`, `UTF-8` encoding, and `LF` line endings. The database dialect is set to `MS SQL`. A system notification for "Breaking news PM Modi arrives..." is visible in the bottom left corner.



# Insert Data

```
1 import { PrismaClient } from "@prisma/client";
2
3 const
4
5 const
6
7 };
8
9 main()
10   .catch((e) => console.error(e))
11   .finally(async () => {
12     await prisma.$disconnect();
13   });
14
```





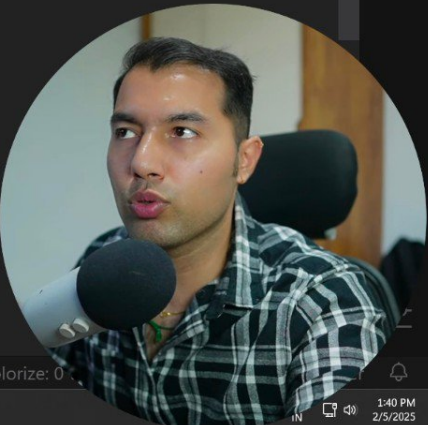
index.js - prisma\_mysql - Visual Studio Code

index.js > ...

```
1 import { PrismaClient } from "@prisma/client";
2
3 const prisma = new PrismaClient();
4
5 main()
6   .catch((e) => console.error(e))
7   .finally(async () => {
8     await prisma.$disconnect();
9   });
10
```

Ln 4, Col 1 Spaces: 4 UTF-8 CRLF {} JavaScript Go Live Colorize: 0

30°C Sunny 1:40 PM 2/5/2025



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5 Min

15 Min

15 Min

Install Prisma Client

Install and generate Prisma Client

To get started with Prisma Client, you need to install the `@prisma/client` package:

```
$ npm install @prisma/client
```

The install command invokes `prisma generate` for you which reads your Prisma schema and generates a version of Prisma Client that works with your schema.

Developer (you)

Prisma CLI

Prisma schema

node\_modules/@prisma/client

1 npm install @prisma/client

2 prisma generate

3 Reads Prisma schema

4 Prisma schema

5 Updates

Thapa Technical

1:37 PM

2/5/2025

# Prisma Client

- Install Prisma Client
  - **npm install @prisma/client**
- Add this in **schema.prisma** above models to enable Prisma Client generation:  
generator client {  
 provider = "prisma-client-js"  
}
- Run **npx prisma generate** to generate Prisma Client after defining
- **npx prisma generate** is automatically executed when running **npx prisma migrate dev**.
- It can also be run manually if the schema is updated outside of migrations.

```
import { PrismaClient } from "@prisma/client";  
  
export const prisma = new PrismaClient();  
  
await prisma.shortLink.findMany();
```



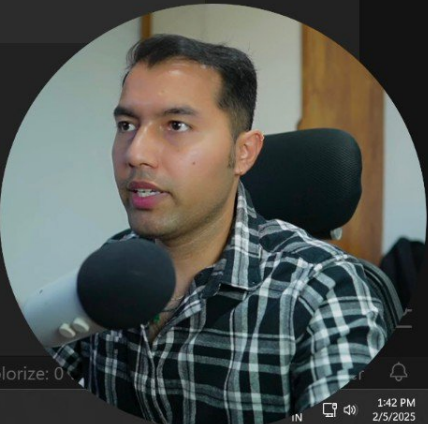
index.js - prisma\_mysql - Visual Studio Code

index.js > main

```
5  const main = async () => {
6    ///? 1 Create (Insert Data)
7    ///? ✓ Single User
8    const user = await prisma.user.create({
9      data: {
10        name: "vinod",
11        email: "thapa@test.com",
12      },
13    });
14    console.log(user);
15  };
16
17  main()
18    .catch((e) => console.error(e))
19    .finally(async () => {
20      await prisma.$disconnect();
21    });
```

Ln 14, Col 20 Spaces: 4 UTF-8 CRLF {} JavaScript Go Live Colorize: 0

30°C Sunny 1:42 PM 2/5/2025





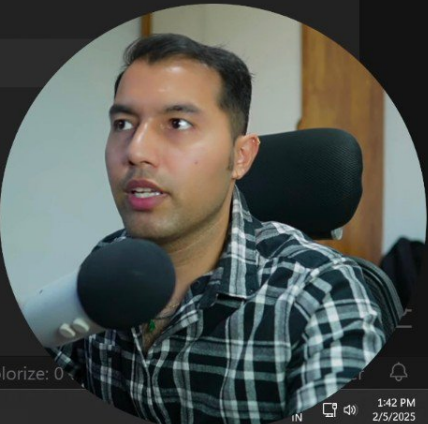
index.js - prisma\_mysql - Visual Studio Code

index.js > ...

```
4
5  const main = async () => {
6    ///1 Create (Insert Data)
7    ///✓ Single User
8    const user = await prisma.user.create({
9      data: {
10        name: "vinod",
11        email: "thapa@test.com",
12      },
13    });
14  };
15
16  main()
17    .catch((e) => console.error(e))
18    .finally(async () => {
19      await prisma.$disconnect();
20    });
```

Ln 14, Col 3 Spaces: 4 UTF-8 CRLF {} JavaScript Go Live Colorize: 0

30°C Sunny 1:42 PM 2/5/2025



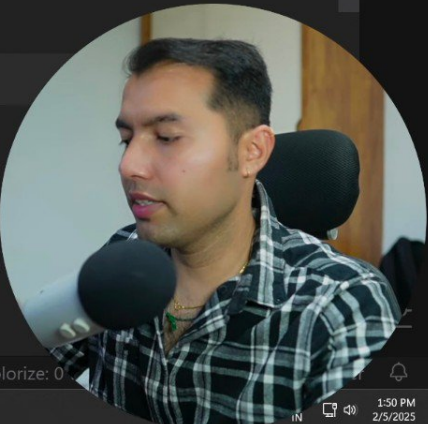
index.js - prisma\_mysql - Visual Studio Code

index.js > main

```
13 // });
14 // console.log(user);
15
16 ///✓ Multiple User
17 const newUsers = await prisma.user.createMany({
18   data: [
19     { name: "Alice", email: "alice@example.com" },
20     { name: "Bob", email: "bob@example.com" },
21   ],
22 });
23 console.log(newUsers);
24 };
25
26 main()
27   .catch((e) => console.error(e))
28   .finally(async () => {
```

Ln 23, Col 23 Spaces: 4 UTF-8 CRLF {} JavaScript Go Live Colorize: 0

1:50 PM 2/5/2025



MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

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Filter objects

- authdb
- mydbprisma
- mysql\_db
- mysql\_users
- prismaql**
- sys
- url\_shortener
- url\_shortener\_mysql
- Tables
  - short\_links
    - Columns
    - Indexes
    - Foreign Keys
    - Triggers
- Views
- Stored Procedures
- Functions

Administration Schemas

Information

Schema: prismaql

Object Info Session

SQL File 1\*

Limit to 1000 rows

```
1 select * from user;
```

| id | email          | name  | createdAt           | updatedAt           |
|----|----------------|-------|---------------------|---------------------|
| 1  | thapa@test.com | vinod | 2025-02-05 08:1 ... | 2025-02-05 08:1 ... |

Result Grid

Filter Rows

Edit

Export/Imports

Wrap Cell Contents

user 2 x

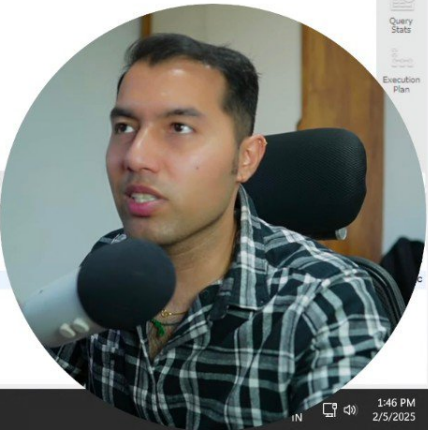
Output

Action Output

| # | Time     | Action                           | Message           |
|---|----------|----------------------------------|-------------------|
| 1 | 13:35:17 | select * from user LIMIT 0, 1000 | 0 row(s) returned |
| 2 | 13:45:41 | select * from user LIMIT 0, 1000 | 1 row(s) returned |

30°C Sunny

1:46 PM 2/5/2025



index.js - prisma\_mysql - Visual Studio Code

index.js > ...

```
4
5 const main = async () => {
6   /// 1 Create (Insert Data)
7   /// ✓ Single User
8   const user = await prisma.user.create({
9     data: {
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

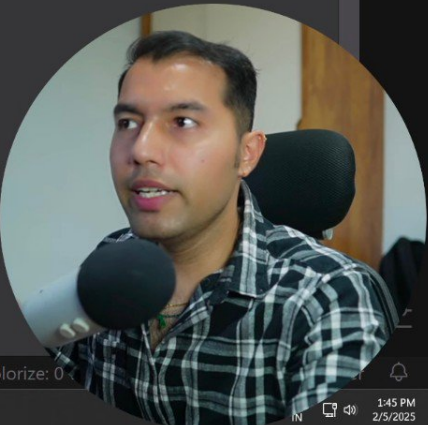
node\_2025\prisma\_project\prisma\_mysql on HEAD (bbf3403) [\$!?] is v1.0.0 via v23.2.0

o) node .\index.js

bbf34035\* 0 0 2 0 5.3.2

Ln 24, Col 1 Spaces: 4 UTF-8 CRLF {} JavaScript Go Live Colorize: 0

30°C Sunny 1:45 PM 2/5/2025





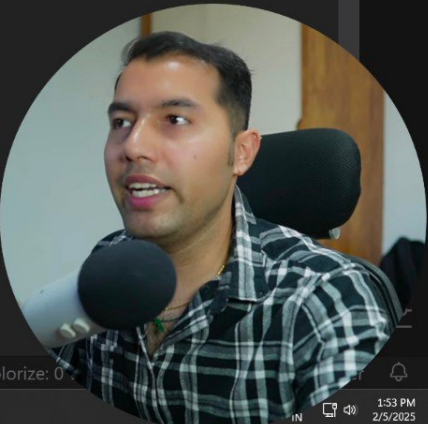
index.js - prisma\_mysql - Visual Studio Code

index.js > main > user

```
23 //? 2 Read (Fetch Data)
24 //? ✓ Get All Users
25 // const users = await prisma.user.findMany();
26 // console.log(users);
27 //? ✓ Get a Single User by ID
28 const user = await prisma.user.findUnique({
29   where: { id: 2 },|
30 });
31 console.log(users);
32 };
33
34 main()
35   .catch((e) => console.error(e))
36   .finally(async () => {
```

Ln 29, Col 22 Spaces: 4 UTF-8 CRLF {} JavaScript Go Live Colorize: 0

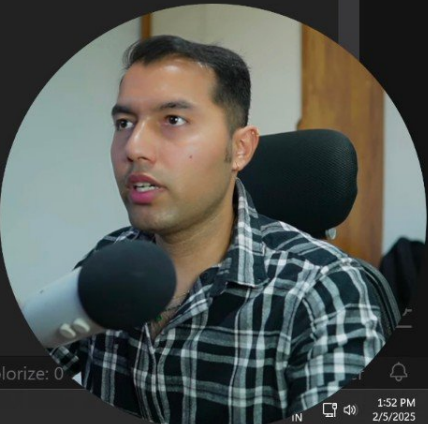
1:53 PM 2/5/2025



Visual Studio Code interface showing a JavaScript file named `index.js` in a project named `prisma_mysql`. The file contains code for fetching data from a database using Prisma.

```
// ...  
21 // });  
22 // console.log(newUsers);  
23 //? 2 Read (Fetch Data)  
24 //? ✓ Get All Users  
25 const users = await prisma.user.findMany();  
26 console.log(users);  
27  
28 };  
29  
30 main()  
31 .catch((e) => console.error(e))  
32 .finally(async () => {  
33   await prisma.$disconnect();  
34 });  
35
```

The code is written in JavaScript and uses Prisma for database operations. It includes comments in Hindi/English. The status bar at the bottom shows the file is at line 26, column 21, with 4 spaces, UTF-8 encoding, and CRLF line endings. The language is JavaScript. The system tray at the bottom shows the date and time as 1:52 PM on 2/3/2025.



# READ DB DATA

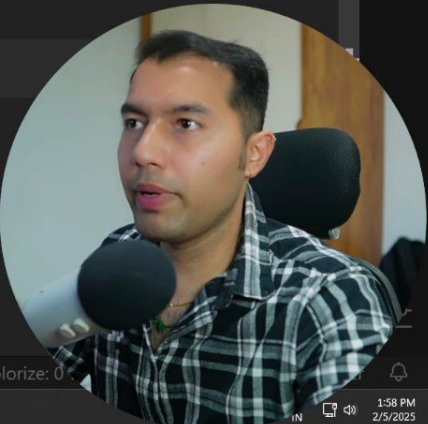
```
13 // });  
14 // console.log(user);  
15  
16  
17  
18  
19  
20 // ],  
21 // });  
22 // console.log(newUsers);  
23 };  
24  
25 main()  
26 .catch((e) => console.error(e))  
27 .finally(async () => {  
28   await prisma.$disconnect();
```



Visual Studio Code interface showing a JavaScript file named `index.js` in the `prisma_mysql` project. The code is currently on line 44, column 3.

```
36 // console.log(user);
37 //? 3 Update (Modify Data)
38 //? ✓ Update One User
39 const updatedUser = await prisma.user.update({
40   where: { id: 3 },
41   data: { name: "BobTheBuilder" },
42 });
43 console.log(updatedUser);
44
45 };
46
47 main()
48 .catch((e) => console.error(e))
49 .finally(async () => {
```

The status bar at the bottom indicates the file is `index.js` in the `prisma_mysql` project, using `JavaScript` syntax, `UTF-8` encoding, and `CRLF` line endings. The system tray shows a temperature of 30°C and the date/time as 1:58 PM on 2/5/2025.



# UPDATE DATA

```
29 //   where: { id: 2 },
30 //   });
31
32
33
34
35 //   });
36 //   console.log(user);
37 //? 3 Update (Modify Data)
38 //? [x] Update One User
39 };
40
41 main()
42   .catch((e) => console.error(e))
```





index.js - prisma\_mysql - Visual Studio Code

index.js > main

```
29 //     where: { id: 2 },
30 //   });
31 //   console.log(user);
32
33 //? ☒ Get Users with Filtering
34 const user = await prisma.user.findMany({
35   where: { name: "vinod" },
36 });
37 console.log(user);
38 };
39
40 main()
41   .catch((e) => console.error(e))
42   finally(async () => {
```

Ln 37, Col 21 Spaces: 4 UTF-8 CRLF {} JavaScript Go Live Colorize: 0

30°C Sunny 1:56 PM 2/5/2025



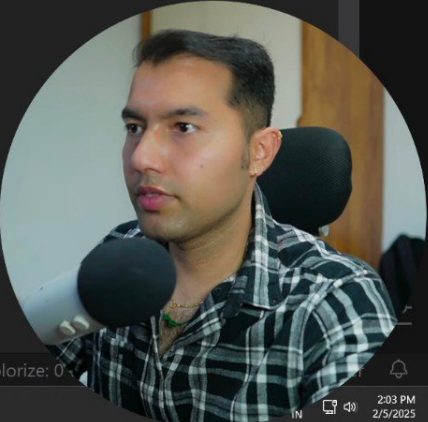
index.js - prisma\_mysql - Visual Studio Code

index.js > main

```
40 //      data: { email: "bob@thapa.com" },
47 //    });
48 //    console.log(updatedUser);
49 //?  Update Multiple Users
50 //    const updatedUser = await prisma.user.updateMany({
51 //      where: { name: "Bob" },
52 //      data: { email: "bob@thapa.com" },
53 //    });
54 //    console.log(updatedUser);
55 //?  Delete (Remove Data)
56 //?  Delete One User
57 const deletedUser = await prisma.user.delete({
58   where: { id: 3 },
59 });
60 console.log(deletedUser);
61 };
62
```

bbf34035\* 0 0 0 3 0 [TypeScript Importer]: Symbols: 0 5.3.2 Spaces: 4 UTF-8 CRLF {} JavaScript Go Live Colorize: 0

30°C Sunny 2:03 PM 2/5/2025



# DELETE DATA

-10

```
index.js > main
// console.log(updatedUser);
45 //   where: { name: "Bob" },
46 //   data: { email: "bob@thapa.com" },
47 // });
48
49
50
51
52 //   data: { email: "bob@thapa.com" },
53 // });
54 // console.log(updatedUser);
55 //? 4 Delete (Remove Data)
56 //? ✓ Delete One User
57 };
58
59 main()
60 .catch((e) => console.error(e))
```



index.js - prisma\_mysql - Visual Studio Code

index.js > main > updatedUser

```
35 // });
36 // console.log(user);
37 //? 3 Update (Modify Data)
38 //? ✓ Update One User
39 const updatedUser = await prisma.user.update({
40   where: { name: "Bob" },
41   data: { email: "bob@thapa.com" },
42 });
43 console.log(updatedUser);
44 //? ✓ Update Multiple Users
45 // const updatedUser = await prisma.user.updateMany({
46 //   where: { id: 3 },
47 //   data: { name: "Bob" },
48 // });
49 // console.log(updatedUser);
50 };
```

bbf34035\* [TypeScript Importer]: Symbols: 0 5.3.2 Spaces: 4 UTF-8 CRLF {} JavaScript Go Live Colorize: 0

30°C Sunny 2:01 PM 2/5/2025





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#### 4. Which databases are supported by Prisma ORM?

Answer:

Prisma supports:

- PostgreSQL
- MySQL
- SQLite
- SQL Server
- MongoDB
- CockroachDB

#### 5. How do you install Prisma in a Node.js project?

Answer:

Run the following commands:

bash

Copy code

```
npm install prisma --save-dev  
npx prisma init
```

This will create a `prisma` folder with a `schema.prisma` file and a `.env` file for your database URL.

#### 6. What is `schema.prisma`?

Answer:

`schema.prisma` is the main configuration file of Prisma.

It defines:

- The database connection ( `datasource` )
- The Prisma client generator
- The data models (tables)

Example:

prisma

Copy code

```
datasource db {  
  provider = "postgresql"  
  url      = env("DATABASE_URL")  
}  
  
generator client {  
  provider = "prisma-client-js"  
}  
  
model User {  
  id    Int    @id @default(autoincrement())  
  name  String  
  email String @unique  
}
```

#### 7. How do you generate the Prisma client?





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Your ankit pawar a fresher give me all interviews questions and answers for the prisma orm

Sure, Ankit 🙌

Here's a complete list of Prisma ORM interview questions and answers — designed especially for freshers like you who want to confidently explain Prisma in interviews (for Full Stack or Backend roles).

## 🧠 Basic Prisma ORM Interview Questions and Answers

### 1. What is Prisma ORM?

Answer:

Prisma is a next-generation Object Relational Mapper (ORM) for Node.js and TypeScript. It helps developers interact with databases easily by auto-generating type-safe queries instead of writing raw SQL.

Example:

ts

Copy code

```
const users = await prisma.user.findMany();
```

This fetches all users from the database — simple and clean!

### 2. What are the main components of Prisma?

Answer:

Prisma has three main components:

1. Prisma Client – Auto-generated and type-safe query builder to access your database.
2. Prisma Migrate – Handles schema changes and database migrations.
3. Prisma Studio – A GUI tool to view and edit your data visually.

### 3. What are the advantages of using Prisma?

Answer:

- ✓ Type safety with TypeScript
- ✓ Easy to use and auto-completion in IDE
- ✓ Better productivity and clean syntax
- ✓ Simplified database migrations
- ✓ Works with multiple databases (PostgreSQL, MySQL, SQLite, MongoDB, etc.)

### 4. Which databases are supported by Prisma ORM?

Answer:

Prisma supports:

- PostgreSQL
- MySQL
- SQLite
- SQL Server

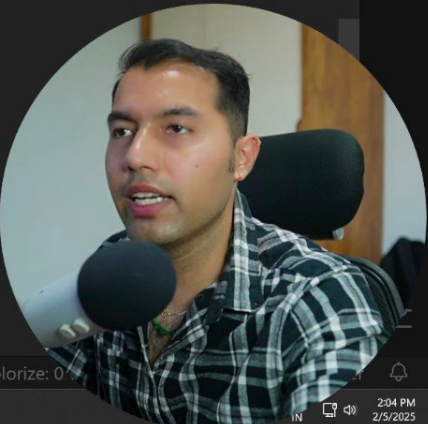
index.js - prisma\_mysql - Visual Studio Code

index.js > main > deletedUsers > where

```
54 // console.log(updatedUser);
55 //? 4 Delete (Remove Data)
56 //? ✓ Delete One User
57 // const deletedUser = await prisma.user.delete({
58 //   where: { id: 3 },
59 // });
60 // console.log(deletedUser);
61 //? ✓ Delete Multiple Users
62 const deletedUsers = await prisma.user.deleteMany({
63   where: [{ id: 1 }, { id: 4 }, { id: 5 }],
64 });
65 console.log(deletedUsers);
66 };
67
68 main()
69   .catch((e) => console.error(e))
70   .finally(async () => {
```

bbf34035\* 0 0 0 3 0 [TypeScript Importer]: Symbols: 0 5.3.2 Spaces: 4 UTF-8 CRLF {} JavaScript Go Live Colorize: 0

30°C Sunny 2:04 PM 2/5/2025





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14. How do you update and delete data using Prisma?

Answer:

✓ Update:

```
ts                                                                    Copy code

await prisma.user.update({
  where: { id: 1 },
  data: { name: "Ankit Pawar" },
});
```

✓ Delete:

```
ts                                                                    Copy code

await prisma.user.delete({
  where: { id: 1 },
});
```

15. How do you handle transactions in Prisma?

Answer:

```
ts                                                                    Copy code

await prisma.$transaction([
  prisma.user.create({ data: { name: "A" } }),
  prisma.post.create({ data: { title: "Post A" } }),
]);
```

Both queries succeed or fail together.

### 🔒 Advanced / Conceptual Questions

16. How does Prisma ensure type safety?

Answer:

Prisma automatically generates a TypeScript client based on your schema.

This means queries like `prisma.user.findMany()` know the exact fields and types at compile-time.

17. What happens when you change the Prisma schema?

Answer:

When you edit `schema.prisma`:

1. Run `npx prisma migrate dev` → updates the database
2. Run `npx prisma generate` → updates the Prisma client

18. What is the `@relation` attribute in Prisma?

Answer:

`@relation` defines relationships between tables (models), like foreign keys in SQL.



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## Intermediate Prisma Questions

11. What is the difference between Prisma Migrate and Prisma Generate?

Answer:

- Prisma Migrate: Creates and applies changes to the database schema.
- Prisma Generate: Generates the Prisma client based on the schema.

12. How do you handle relations in Prisma?

Answer:

You can define one-to-many, many-to-many, and one-to-one relationships.

Example (one-to-many):

prisma

Copy code

```
model User {
  id      Int      @id @default(autoincrement())
  name    String
  posts   Post[]
}

model Post {
  id      Int      @id @default(autoincrement())
  title   String
  author  User?    @relation(fields: [authorId], references: [id])
  authorId Int?
}
```

13. How do you perform filtering or conditions in Prisma?

Answer:

ts

Copy code

```
const users = await prisma.user.findMany({
  where: { name: { contains: "Ankit" } },
});
```

14. How do you update and delete data using Prisma?

Answer:

✓ Update:

ts

Copy code

```
await prisma.user.update({
  where: { id: 1 },
  data: { name: "Ankit Pawar" },
});
```

✓ Delete:

ts

Copy code



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7. How do you generate the Prisma client?

Answer:

After defining your models, run:

```
bash
npx prisma generate
```

Copy code

This creates the Prisma Client that you can import and use in your code.

8. How do you run Prisma migrations?

Answer:

Run:

```
bash
npx prisma migrate dev --name init
```

Copy code

This applies schema changes to your database and updates the migration history.

9. How do you use Prisma Client in a project?

Answer:

```
ts
import { PrismaClient } from '@prisma/client';
const prisma = new PrismaClient();

async function main() {
  const allUsers = await prisma.user.findMany();
  console.log(allUsers);
}

main();
```

Copy code

10. What is Prisma Studio used for?

Answer:

Prisma Studio is a visual database explorer — it lets you view, edit, and delete data directly in your browser.

To open it:

```
bash
npx prisma studio
```

Copy code



## Intermediate Prisma Questions

11. What is the difference between Prisma Migrate and Prisma Generate?

Answer:

- Prisma Migrate: Creates and applies changes to the database schema.



**Chetanya sharma** ✓ • 3rd+[+ Follow](#)

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3mo • 🌐

✓ 🎯 **15 Basic Prisma Interview Questions Every Backend Developer Should Know (With Answers)**

📌 **1. What is Prisma in Node.js?**

**Answer:** Prisma is an open-source next-generation ORM for Node.js and TypeScript that helps developers write type-safe database queries using an auto-generated query builder.

📌 **2. Which databases does Prisma support?**

**Answer:** Prisma supports PostgreSQL, MySQL, SQLite, SQL Server, CockroachDB, and MongoDB.

📌 **3. What is the schema.prisma file?**

**Answer:** It's the central configuration file that defines your database models, datasource (DB connection), and generator (like Prisma Client).

📌 **4. How do you generate Prisma Client?**

**Answer:** By running `npx prisma generate`. It reads `schema.prisma` and generates a fully-typed Prisma Client.

📌 **5. How to migrate your schema to the database?**

**Answer:** Use `npx prisma migrate dev --name init` to create and apply a new migration based on your schema changes.

📌 **6. What is Prisma Client?**

**Answer:** A type-safe query builder auto-generated from your Prisma schema to interact with your database using JavaScript/TypeScript.

📌 **7. How do you fetch all users using Prisma Client?**

**Answer:** `const users = await prisma.user.findMany();`



Add a comment...



I chose Prisma because it makes database handling easier, type-safe, and faster.

It's beginner-friendly and integrates well with frameworks like Next.js and Express.js.

## 22. How is Prisma better than traditional ORMs like Sequelize?

Answer:

- Prisma provides auto-generated, type-safe queries
- Easier migrations
- Modern syntax
- Better integration with TypeScript

## 23. How can you connect Prisma with your PostgreSQL database?

Answer:

Edit `.env`:

```
ini
```

[Copy code](#)

```
DATABASE_URL="postgresql://user:password@localhost:5432/mydb"
```

Then run:

```
bash
```

[Copy code](#)

```
npx prisma migrate dev
```

## 24. Can Prisma be used in Next.js projects?

Answer:

Yes, Prisma integrates perfectly with Next.js for building full-stack apps — often used with APIs and server actions.

## 25. What are some limitations of Prisma?

Answer:

- Not ideal for very complex raw SQL queries
- Limited database driver support compared to older ORMs
- Slightly heavier initial setup

Would you like me to give you a short version (20 must-know Q&A) for *quick revision before interviews* or a Prisma + Next.js interview combo Q&A (many companies ask both together)?



+ Ask anything





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18. What is the `@relation` attribute in Prisma?

Answer:

`@relation` defines relationships between tables (models), like foreign keys in SQL.

19. Can Prisma be used with MongoDB?

Answer:

Yes. Prisma supports MongoDB, but its syntax is slightly different since MongoDB is NoSQL.

20. How to seed initial data in Prisma?

Answer:

You can create a `prisma/seed.ts` file and run:

```
bash
```

[Copy code](#)

```
npx prisma db seed
```

Example:

```
ts
```

[Copy code](#)

```
import { PrismaClient } from '@prisma/client';  
const prisma = new PrismaClient();  
  
async function main() {  
  await prisma.user.create({ data: { name: "Ankit", email: "ankit@example.com" } });  
}  
  
main();
```

🗨️ Bonus HR-Type Questions for a Fresher

21. Why did you choose Prisma ORM?

Answer:

I chose Prisma because it makes database handling easier, type-safe, and faster.

It's beginner-friendly and integrates well with frameworks like Next.js and Express.js.

22. How is Prisma better than traditional ORMs like Sequelize?

Answer:

- Prisma provides auto-generated, type-safe queries
- Easier migrations
- Modern syntax
- Better integration with TypeScript

23. How can you connect Prisma with your PostgreSQL database?

Answer:



and apply a new migration based on your schema changes.

#### 6. What is Prisma Client?

Answer: A type-safe query builder auto-generated from your Prisma schema to interact with your database using JavaScript/TypeScript.

#### 7. How do you fetch all users using Prisma Client?

Answer: `const users = await prisma.user.findMany();`

#### 8. How do you filter records in Prisma?

Answer: Use the where clause.

```
const activeUsers = await prisma.user.findMany({
  where: { status: 'active' }
});
```

#### 9. How do you relate models in Prisma?

Answer: By defining relationships in the schema like this:

```
model Post {
  id Int @id @default(autoincrement())
  title String
  author User @relation(fields: [authorId], references: [id])
  authorId Int
}
```

#### 10. How to create a user using Prisma?

Answer: `await prisma.user.create({`

```
  data: {
    name: 'Chetan Sharma',
    email: 'chetan@example.com',
  },
});
```

#### 11. How do you update a record in Prisma?



Add a comment...





,,

 11. How do you update a record in Prisma?

Answer: `await prisma.user.update({  
 where: { id: 1 },  
 data: { name: 'Updated Name' },  
});`

 12. How to delete a record using Prisma?

Answer: `await prisma.user.delete({  
 where: { id: 1 },  
});`

 13. What is prisma studio?

Answer: Prisma Studio is a GUI for your database. You can run it using `npx prisma studio`.

 14. How do you handle nested writes (create with relations)?

`await prisma.user.create({  
 data: {  
 name: 'John',  
 posts: {  
 create: [{ title: 'First Post' }],  
 },  
 },  
});`

 15. What are @id, @default, and @relation in Prisma?

Answer:

@id: Marks a field as primary key.

@default: Sets a default value.

@relation: Defines relations between models.



Add a comment...

